

Computer Science & IT

Discrete Mathematics



Set Theory & Algebra

Lecture No. 02



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Recap of Previous Lecture



Topic

Sets and representation of sets



Topic

Types of sets



Topic

Number of subsets of a given set



Topics to be Covered



Topic

Number of subsets of a given set



Topic

Power set



Topic

Venn diagram



Topic

Set operations and properties of set operations



$$(a+b)^n = nC_0(a)^0(b)^n + nC_1(a)^1(b)^{n-1} + nC_2(a)^2(b)^{n-2} + \dots + nC_n(a)^n(b)^0$$



Topic : Number of subsets of a set 'A' of cardinality 'n'

Let $|A| = n$,

- ① Total no. of subsets of set A $= {}^nC_0 + {}^nC_1 + \dots + {}^nC_n = 2^n$
- ② Number of subsets of set A, such that size of each subset is exactly 'k' $= {}^nC_k$ {where $k \leq n$ }
- ③ Number of subsets of set A, such that size of each subset is at most 'k' $= {}^nC_0 + {}^nC_1 + \dots + {}^nC_k$
- ④ Number of subsets of set A, such that size of each subset is at least 'k' $= {}^nC_k + {}^nC_{k+1} + \dots + {}^nC_n$

Note ∴ If $|A| = n$, then

number of proper subsets of set $A = \underbrace{2^n}_{\text{all subsets}} - \underbrace{1}_{\text{except set A itself.}}$

Another way of thinking,

$$\text{Let } A = \{1, 2, 3, 4, \dots, n\}$$

then,

Number of
Subsets of set $A =$

$$\underbrace{2 * 2 * 2 * 2 * \dots * 2}_{n \text{ times } '2'} = 2^n$$

Q: Let $|A| = 5$,

How many subsets of set A are there of size at most '3'.

Soluⁿ.

$$\text{Number of subsets of size at most '3'} = {}^5C_0 + {}^5C_1 + {}^5C_2 + {}^5C_3$$

$$= 1 + 5 + 10 + 10$$

$$= \textcircled{26}$$

Q: Let $A = \{0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10\}$

How many subsets of set A are there that contains exactly three odd numbers.

Soln. $= {}^5C_3 \times 2^6 = 10 \times 64 = 640$

$\uparrow$ Choose exactly three out of '5' odd elements

$\uparrow$ And

$\uparrow$ 2^6 different choices w.r. even elements

* Concept w.r.t. number of Supersets:-

Number of supersets of a given set will always be defined w.r.t. Universal set.

Q. Let $A = \{1, 2, 4, 5\}$ is a set derived w.r.t universal set $U = \{1, 2, 3, 4, 5, 6, 7, 8, 9, 10\}$
How many supersets of set A are there?

Soln.

$$U = \{1, 2, 3, 4, 5, 6, 7, 8, 9, 10\}$$

Number of Super-Sets of set A = $1 \times 1 \times 2 \times 1 \times 1 \times 2 \times 2 \times 2 \times 2 \times 2 = 2^6 = 64$

Each element of set A must be present in the Superset of set A.

For every other element we have '2' choices
Either it will be present or it will be absent

Ans

64

Note :-

We know,

$${}^nC_r = {}^nC_{n-r}$$

$$\begin{array}{ccccccc} \downarrow & & \downarrow & & \downarrow & & \downarrow \\ {}^nC_0 & + & {}^nC_1 & + & {}^nC_2 & + & \dots + {}^nC_{n-2} + {}^nC_{n-1} + {}^nC_n \end{array}$$

$$\therefore \underbrace{{}^nC_0 + {}^nC_1 + \dots + {}^nC_k}_{\text{Number of subsets of size at most 'k'}} = \underbrace{{}^nC_{n-k} + {}^nC_{n-k+1} + \dots + {}^nC_n}_{\text{Number of subsets of size at least 'n-k'}}$$

Number of subsets of size at most 'k'

=

Number of subsets of size at least "n-k"



Topic : Power Set

Power Set:

Let 'A' is any finite set, Power set of set 'A' can be denoted by $P(A)$ or 2^A , and it is defined as.

$P(A)$ = Set of all subsets of set A.

$P(A) = \{ \underbrace{\hspace{10em}}_{\text{Elements will be all subsets of set A.}} \}$

Q:- Let $A = \{1, 2, 3\}$, then

$$P(A) = ?$$

Soluⁿ $P(A) = \{ \emptyset, \{1\}, \{2\}, \{3\}, \{1, 2\}, \{1, 3\}, \{2, 3\}, \{1, 2, 3\} \}$

$$= \{ \{ \}, \{1\}, \{2\}, \{3\}, \{1, 2\}, \{1, 3\}, \{2, 3\}, \{1, 2, 3\} \}$$

Note:- ① $\emptyset$ is a subset of every set.

∴ $\emptyset$ is a subset of $P(A)$ as well
for any set A .

② $\emptyset$ is always an element of $P(A)$
for any set A .



Topic : Cardinality of Power Set

Cardinality of $P(A)$ = $|P(A)|$ = Number of elements in $P(A)$
= Number of subsets of set A
= $2^{|A|}$

$$|P(A)| = 2^{|A|}$$

Q: Let $\emptyset$ be an empty set, then

$$|P(P(\emptyset))| = ?$$

- (a) 0
- (b) 1
- ☒ (c) 2
- (d) Can not be defined.

$$|\emptyset| = 0$$

$$|P(\emptyset)| = 2^{|\emptyset|} = 2^0 = 1$$

$$|P(P(\emptyset))| = 2^{|P(\emptyset)|} = 2^1 = 2$$

Q: Let $\emptyset$ be an empty set, then

$$|P(P(\emptyset))| = ?$$

* $\emptyset = \{ \}$

* $P(\emptyset) = \{ \underbrace{\emptyset} \} = \{ \{ \} \}$

$$\{ \{ \} \} \neq \{ \}$$

$$|P(\emptyset)| = |\{ \{ \} \}| = 1$$

* $P(P(\emptyset)) = \{ \underbrace{\emptyset}_{\text{Subset of size } = 0}, \underbrace{\{ \emptyset \}}_{\text{Subset of size } = 1} \} = \{ \{ \}, \{ \{ \} \} \}$

Subset of size = 0

elements of the set will be subset of set $P(\emptyset)$ i.e., subsets of $\{ \{ \} \}$

Subset of size = 1

2 elements in the set

$$\therefore |P(P(\emptyset))| = 2$$

Q: Let $A = \{1, 2, 3, \{1, 2\}, \{\{2, 3\}\}\}$ $|A| = 5$

Which of the following is/are true

✓ (a) $1 \in A$

✗ (b) $\{1\} \in A$

✗ (c) $1 \subseteq A$

✓ (d) $\{1\} \subseteq A$

↑
element from
set 'A'

✓ (e) $\{1, 2\} \in A$

✓ (f) $\{1, 2\} \subseteq A$

(g) $\{\{1, 2\}\} \in A$

✓ (h) $\{\{1, 2\}\} \subseteq A$

(i) $\{2, 3\} \in A$

✓ (j) $\{2, 3\} \subseteq A$

✓ (k) $\{\{2, 3\}\} \in A$

(l) $\{\{2, 3\}\} \subseteq A$

✓ (m) $\{\{\{2, 3\}\}\} \subseteq A$

(n) $\emptyset \in A$

✓ (y) $\emptyset \in P(A)$

✓ (x) $\emptyset \subseteq A$

✓ (z) $\emptyset \subseteq P(A)$

$$B = \{1, 2, 3\}$$

$$1 \in B \quad \checkmark$$

$$1 \subseteq B \quad \times$$

$$\{1\} \in B \quad \times$$

$$\{ \quad \} \subseteq B$$

elements must be
'0' or more elements
from set B

Q1 For a set A , the power set of A is denoted by 2^A . If $A = \{5, \{6\}, \{7\}\}$ which of the following options are true?

1. $\emptyset \in 2^A$
2. $\emptyset \subseteq 2^A$
3. $\{5, \{6\}\} \in 2^A$
4. $\{5, \{6\}\} \subseteq 2^A$

H.W.

Q2. Let $P(S)$ denote the power set of a set S . Which of the following is always true?

A. $P(P(S)) = P(S)$

B. $P(S) \cap P(P(S)) = \{\emptyset\}$

C. $P(S) \cap S = P(S)$

D. $S \notin P(S)$



Topic : Venn Diagram

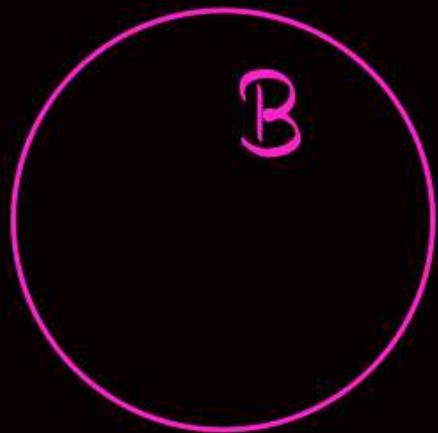
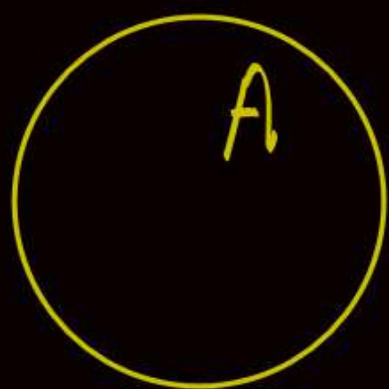
Venn diagram is used to represent the relationship among the sets pictorially.

$$B \subseteq A$$

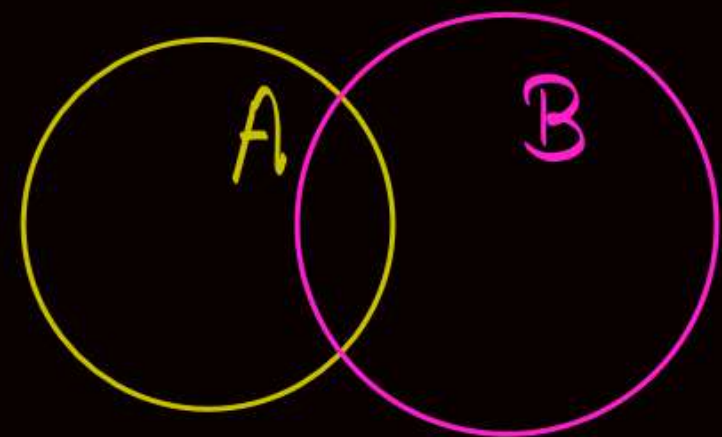
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$$\Rightarrow \begin{aligned} A \cap B &= B \\ A \cup B &= A \end{aligned}$$



No shared region between A & B.
then $A \cap B = \emptyset$



$$A \cap B \neq \emptyset$$

i.e. there are some
Common elements
between set A
and set B



Topic : Set Operations



- ✓ ☐ Complement of a set
- ✓ ☐ Union of two sets
- ✓ ☐ Intersection of two sets
- ✓ ☐ Set difference
- ✓ ☐ Symmetric difference of two sets



Topic : Complement of a set

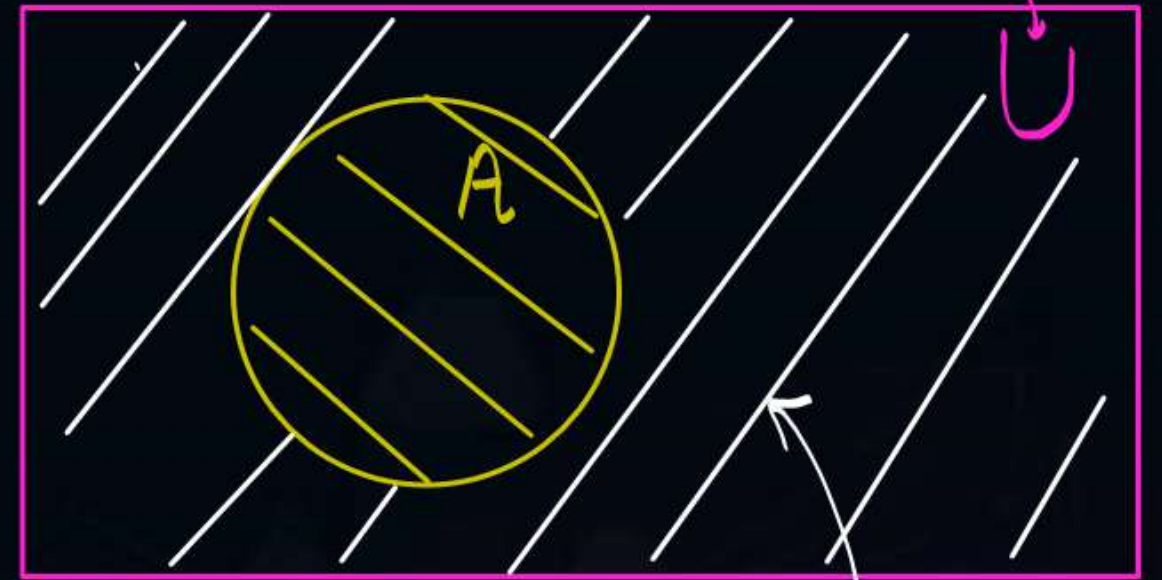
Complement of set is always defined w.r.t. universal set.

$$A^c = \{x \mid x \in U \text{ and } x \notin A\}$$

$$U = \{1, 2, 3, 4, 5\}$$

$$A = \{1, 4\}$$

$$A^c = \{2, 3, 5\}$$



A^c

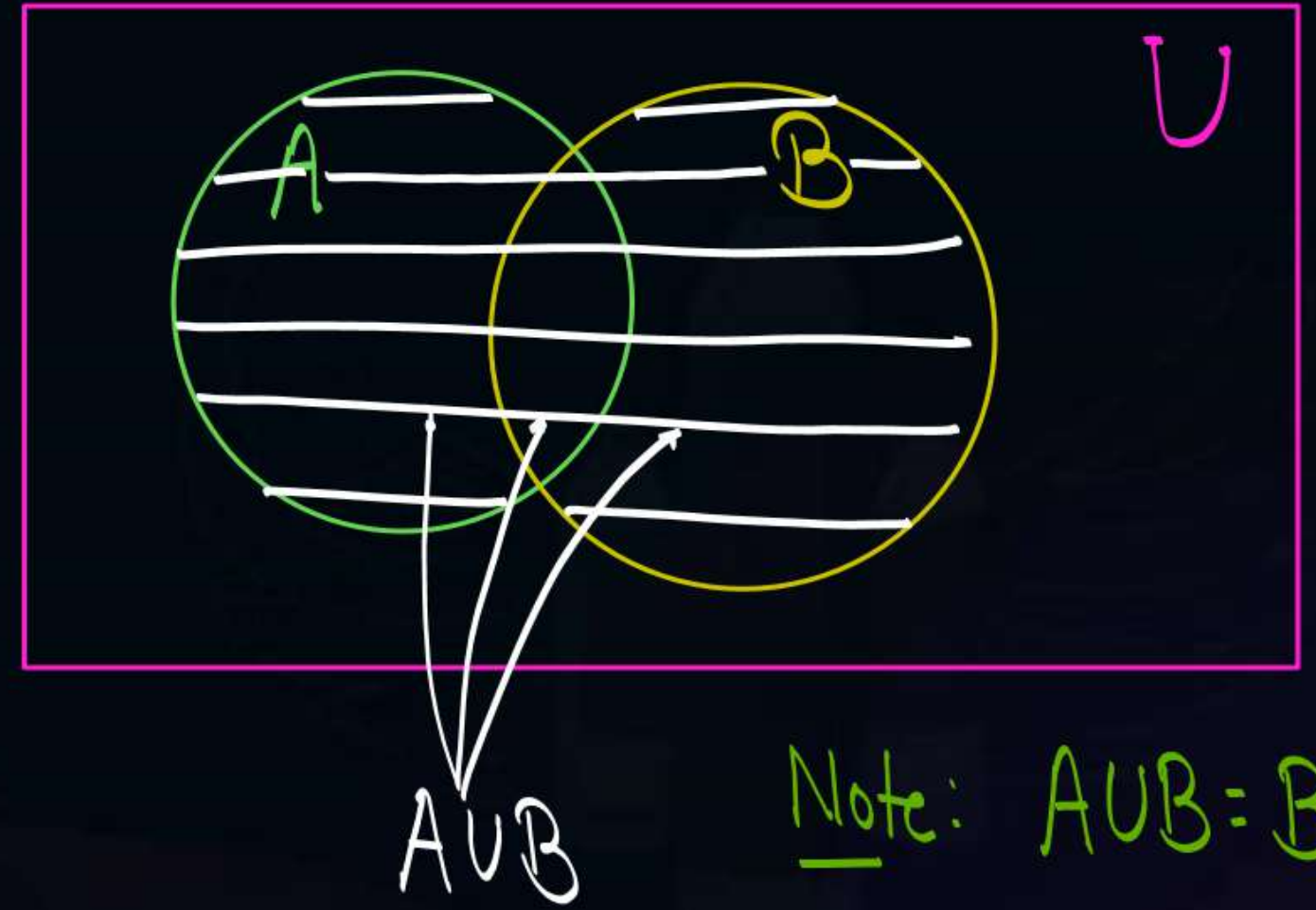


Topic : Union of two sets

$$A \cup B = \{x \mid x \in A \text{ or } x \in B\}$$

Note

$$A \cup A^c = U$$



Note: $A \cup B = B \cup A$



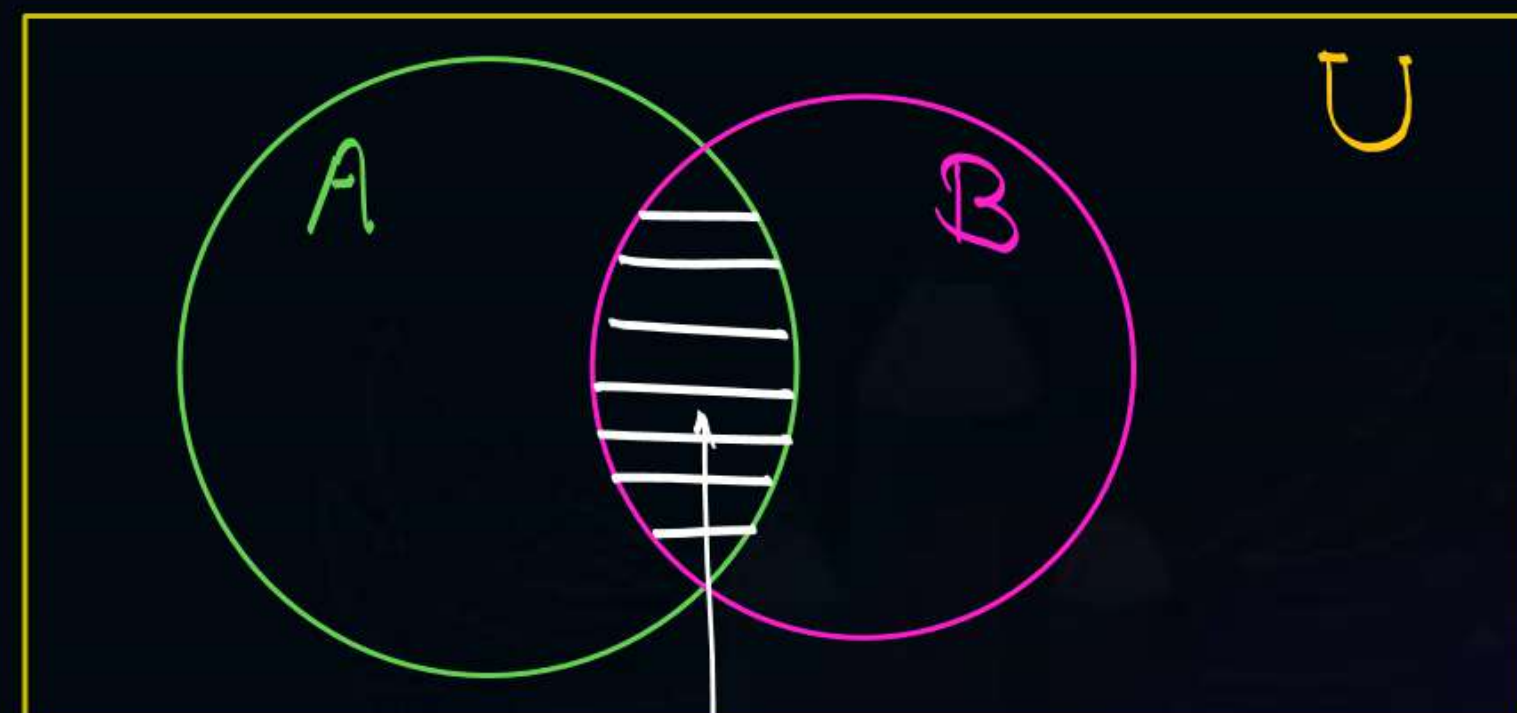
Topic : Intersection of two sets

$$A \cap B = \{x \mid x \in A \text{ and } x \in B\}$$

Note:

$$\text{If } A \cap B = \emptyset.$$

then set A and set B
are called dis-joint sets



$A \cap B$

Note:

$$A \cap B = B \cap A$$



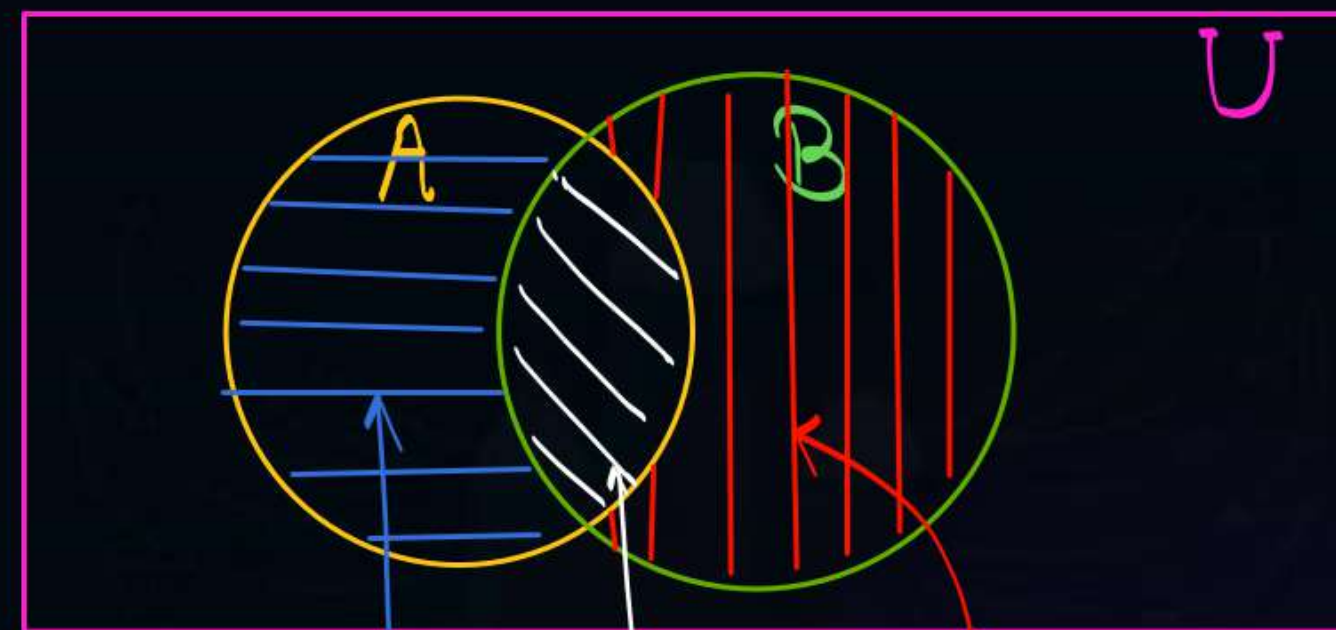
Topic : Set difference

let A & B are two sets,

$$A-B = \{x \mid x \in A \text{ and } x \notin B\}$$

$$B-A = \{x \mid x \in B \text{ and } x \notin A\}$$

" $A-B$ " will contain the elements that are present in set A , but not present in set B



Note:- $A-B = B-A$

if and only if $A = B$.

And in that case $A-B = B-A = \{\} = \emptyset$

$$A-B = (A - (A \cap B)) \quad B-A = (B - (A \cap B))$$

Note: in general $A-B \neq B-A$



2 mins Summary



✓ **Topic**

Power set

✓ **Topic**

Venn diagram

Topic

Set operations and properties of set operations

Next class

time / Vishal Rawtiyapu
↑
Capital



THANK - YOU